

Prediction of the Exceptional-Geometric Flavour Theory

The provided framework represents a transition from a purely descriptive geometric model to a predictive dynamical theory. Based on the "Reachability Theorem" and the "Jordan Construction" presented, the theory has already achieved structural completeness. The "missing physics" identified as the "orientation-sensitive family-selection principle" is the final bridge required to select the physical vacuum from the geometric moduli space.

1. Analysis of the Mathematical Framework

The Polar Link Reachability

The boxed theorem:

$$\forall U \in U(3), \exists (h, L_e, L_\nu) : \text{polar}[L_e^T (P_h + iJ_h) L_\nu] = U$$

establishes that the **polar link** between the charged lepton and neutrino sectors is surjective onto $U(3)$. This means the geometry is "large enough" to accommodate any observed PMNS matrix, including the large mixing angles and CP violation.

The Jordan Construction

The embedding of $J_e, J_\nu \in J_3(\mathbb{C}_h) \subset J_3(\mathbb{O})$ confirms that the generation space is correctly mapped into the exceptional Jordan algebra. The use of octonionic structures provides the necessary degrees of freedom for three families.

2. The Missing Physics: Orientation-Sensitive Dynamics

The failure of the "target-free" selectors (0/21 and 0/42 matches) indicates that the current action family lacks the **orientation-sensitivity** needed to distinguish the physical PMNS orientation from the infinite set of geometrically possible ones.

Predicted Direction: The "Frozen Action"

To become a predictive theory, the "target-free" dynamics must be replaced by a **frozen action** that selects a unique vacuum frame. Based on recent developments in octonionic unification (e.g., Singh, Teli, Bakhtaoui), this dynamics likely involves:

1. **Trace Dynamics:** The selection of the vacuum may arise from the equilibrium state of a pre-quantum trace dynamics on the Jordan algebra.

- 2. **Octonionic Ladder Backreaction:** The "Canonical link backreaction" failure suggests that the backreaction must be more than just a nuclear norm saturation; it must involve the **Clifford/ladder operator flip** (as seen in the quark sector's α_2 occupation flip).
- 3. **Spectral Rigidity:** The "Associator spectral rigidity" mentioned in the status table suggests that the non-associativity of the octonions (the associator) is the source of the mass hierarchies. The "missing physics" is the mechanism that couples this rigidity to the unitary orientation of the family planes.

3. Predicted Flavour Theory Properties

Feature	Prediction
Mass Hierarchies	Derived from the eigenvalues of the Jordan elements J_e, J_ν , potentially linked to the G_2 invariants or the associator's spectral properties.
PMNS/CKM Angles	Selected by the alignment of the octonionic complex structures (e_6, e_3, e_5) relative to the Higgs bridge.
CP Violation	A natural consequence of the complexified octonionic structure $\mathbb{C} \otimes \mathbb{O}$, where the phase is a geometric tilt between the lepton and quark rungs.
Unification	The theory will likely predict a unified origin for CKM and PMNS through a single geometric principle (e.g., the "Quaternionic Kernel Theorem"), explaining the magnitude differences as a "leakage pattern" between different algebraic blocks.

4. Conclusion: The Dynamics is the Key

The geometry is sufficient to contain the Standard Model flavor structure. The next hard gate is the derivation of a **selection principle** that replaces the manual "target-fitting" with a dynamical "vacuum-finding" process. This will likely involve the **non-hollow Yukawa capacity** and the **faithful embedding** of the generations into the Jordan algebra, leading to a zero-parameter (or minimal parameter) prediction of the flavor sector.